

Ayush Rai

Full-Stack Web Engineer

Computer Science student focused on full-stack web development, modern frontend engineering, AI-powered applications and open-source development. Interested in building practical products and learning new technologies through real-world projects.

ayushraistudio@gmail.com GitHub LinkedIn Portfolio

EDUCATION

Raj Kumar Goel Institute of Technology (RKGIT)

Expected 2028

B.Tech in Computer Science

Delhi, India

TECHNICAL SKILLS

- **Frontend:** React.js, Next.js, HTML5, CSS3, JavaScript, TypeScript, Tailwind CSS, Vite
- **Backend & APIs:** Node.js, Express.js, FastAPI, REST APIs, Netlify Functions
- **Databases & Cloud:** MongoDB, Firebase, SQLite, Vercel, Netlify, Cloudflare Tunnel
- **AI / ML:** Python, XGBoost, ConvLSTM, PyTorch/TensorFlow concepts, NumPy, Machine Learning
- **Tools:** Git, GitHub, GitHub Actions, Docker, Postman, VS Code

PROJECTS

Love Letter Service

Live Demo ↗

HTML, CSS, JavaScript, Netlify Functions, GitHub

- Developed a multi-page web service for creating personalized digital love letters.
- Implemented recipient details, photo uploads, AI flow, custum canvas and E2EE 3D letter.
- Used Netlify Functions for backend processing and email-related workflows.
- Deployed and maintained the project using GitHub and Netlify.

GitHub Contribution Showcase API

Live Demo ↗

Next.js, TypeScript, GitHub GraphQL API, SVG, Vercel

- Built a GitHub GraphQL-based API for displaying real user contribution statistics.
- Implemented accurate contribution streak, active days, and total contribution calculations.
- Generated clean SVG-based widgets for presenting GitHub activity data.
- Deployed the API and interactive widgets using Vercel.

react-behavior-guard

Live Demo ↗

React, npm, Open-Source

- Authored a custom React hook for tracking user behavioral patterns during online test sessions.
- Designed the library around detecting and handling potentially suspicious browser-session behaviors.
- Published the package to npm as an open-source developer tool.

AI Krishi Advisor

Live Demo ↗

React, Python, Machine Learning

- Developed an AI-based agriculture application designed to assist users with crop and plant-related information.
- Worked with image-based plant disease identification concepts and machine-learning inference workflows.
- Integrated a modern web frontend with an AI/ML backend workflow.

AIML-Based Nowcasting

Live Demo ↗

Next.js, FastAPI, XGBoost, ConvLSTM

- Built an AIML-based nowcasting system for thunderstorm and lightning risk using atmospheric observations and radar data.
- Used XGBoost to analyse atmospheric conditions such as temperature, pressure, humidity and wind speed.
- Used ConvLSTM to analyse radar-image sequences and storm evolution.
- Combined model outputs through a fusion layer to calculate a final risk probability.
- Integrated Open-Meteo weather telemetry and RainViewer radar data with a FastAPI backend and interactive map frontend.

Exam Portal

Live Demo ↗

React, Firebase

- Developed a secure online MCQ examination portal for conducting online tests.
- Integrated Firebase for real-time database management and user authentication.
- Designed an intuitive user interface for seamless test-taking experience.

CERTIFICATIONS & LEARNING

- **AWS Solutions Architecture** — AWS / Forage
  Completed practical tasks involving designing a simple, scalable hosting architecture. (Oct 12, 2025)
- **Generative AI & Agents** — Microsoft
  Successfully completed Introduction to Generative AI and Agents. (Oct 12, 2025)
- **Java Full Stack Developer** — AICTE / EduSkills
  Completed a 10-week Java Full Stack Developer Virtual Internship. (Oct–Dec 2025)
- **Full Stack Web Development** — Cursa
  Completed Full Stack Web Development course by WB Web Development Solutions. (Oct 30, 2025)
- **Power BI** — Microsoft
  Successfully completed Choose a Power BI model framework. (Oct 12, 2025)
- **JPMorgan Software Engineering** — JPMorgan Chase & Co. / Forage
  Completed practical tasks including Project Setup, Kafka Integration, H2 Integration and REST API Integration. (Oct 19, 2025)
- **Oracle Cloud Data Science Professional** — Oracle University
  Earned Oracle Cloud Infrastructure 2025 Certified Data Science Professional certification. (Oct 13, 2025)
- **Oracle AI Foundations Associate** — Oracle University
  Earned Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate certification. (Oct 11, 2025)
- **Python Essentials 1** — Cisco Networking Academy / Python Institute
  Completed Python Essentials 1, covering Python programming fundamentals and problem solving. (Oct 29, 2025)
- **AWS Cloud Practitioner CLF-C02** — Udemy
  Completed [NEW] Ultimate AWS Certified Cloud Practitioner CLF-C02 2026 course. (Jun 26, 2026)
- **Tata GenAI Powered Data Analytics** — Tata / Forage
  Completed practical tasks in data analysis, risk profiling, AI-based delinquency prediction and collections strategy. (Oct 12, 2025)

ACHIEVEMENTS & ACTIVITIES

- **Smart-Site Sprint – 2nd Position**: Secured 2nd position in Designing domain in the intra-college technical event “Smart-Site Sprint”. (Dec 14–15, 2024)
- **Brawlbit 2.0 – Volunteer**: Volunteered in BRAWLBIT'S 2.0, organized by the CS Department. (Sep 10, 2025)
- **RIFT '26 Hackathon**: Participated in RIFT '26 Hackathon representing RKGIT. (2026)

ADDITIONAL INTERESTS

Web design, modern UI/UX, full-stack development, artificial intelligence, machine learning, cloud technologies, open-source development and building practical software products.